

Hybrid Movie Recommendation System - Project Documentation

1. Project Overview

The Hybrid Movie Recommendation System is a personalized recommendation engine developed using machine learning techniques to suggest relevant movies to users based on their historical behavior and movie preferences.

The system combines Content-Based Filtering and Collaborative Filtering approaches to improve recommendation quality while addressing limitations associated with individual recommendation techniques.

2. Business Objective

Streaming platforms contain thousands of movies, making content discovery difficult for users.

The objective of this project is to:

- Personalize movie recommendations.
 - Improve user engagement.
 - Increase content consumption.
 - Enhance user retention.
 - Address the cold-start problem for new users.
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3. Dataset

Dataset: MovieLens Latest Small Dataset

Files Used:

- movies.csv
- ratings.csv
- tags.csv

Key Attributes:

- User ID
- Movie ID
- Rating
- Genres
- Tags

4. Methodology

Data Cleaning

- Missing value handling
- Duplicate removal
- Data consistency validation

Exploratory Data Analysis

- Rating distribution analysis
- Most rated movies
- Most active users
- User-item sparsity analysis

Content-Based Filtering

Techniques Used:

- TF-IDF Vectorization
- Cosine Similarity

Features:

- Genres
- User-generated tags

Output:

- Movie similarity scores

Collaborative Filtering

Algorithm:

- Singular Value Decomposition (SVD)

Purpose:

- Learn latent user and movie factors.
- Predict ratings for unseen movies.

Hybrid Recommendation System

Formula:

Hybrid Score =
 $0.4 \times \text{Content Score} +$
 $0.6 \times \text{Collaborative Score}$

Benefits:

- Improved personalization
- Better recommendation quality
- Reduced weaknesses of individual models

Cold Start Handling

For new users:

- Popular movies
 - Highly-rated movies
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5. Model Evaluation

Performance Metrics:

RMSE: 0.8807

Precision@10: 0.5660

Recall@10: 0.2794

Interpretation:

- Good prediction accuracy.
 - More than half of recommended movies are relevant.
 - Effective retrieval of relevant content.
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6. Technologies Used

- Python
 - Pandas
 - NumPy
 - Scikit-Learn
 - Surprise Library
 - Streamlit
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7. Deployment Strategy

- Models serialized using Pickle.
 - Streamlit application for user interaction.
 - GitHub repository for version control.
 - Cloud deployment using Streamlit Community Cloud.
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8. Future Enhancements

- Neural Collaborative Filtering
 - Deep Learning Recommendation Models
 - Real-Time User Feedback Integration
 - Explainable Recommendations
 - Cloud-Native Deployment
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9. Conclusion

The Hybrid Movie Recommendation System successfully combines content-based and collaborative filtering techniques to generate personalized movie recommendations. The system demonstrates strong recommendation performance and provides a scalable foundation for deployment on streaming platforms.